Metallic Waveguides

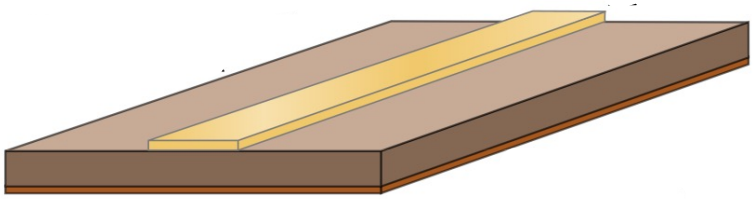
The following subjects will be covered

- Introduction.
- The Metallic Rectangular Waveguide.
- TE and TM Modes.
- Examples.

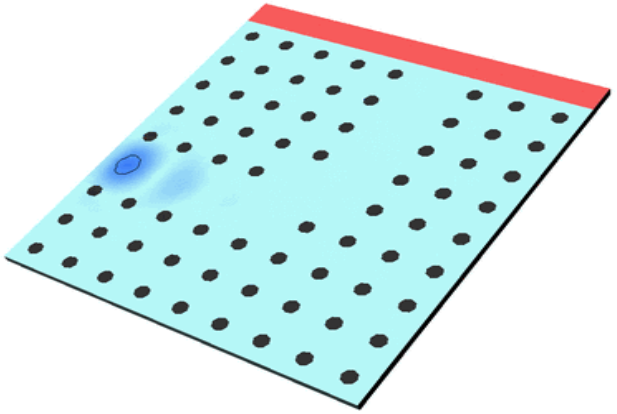
Introduction: Waveguiding

High energy and high frequency (microwave) EM waves can be transported for short distances using "waveguides" made of metal.





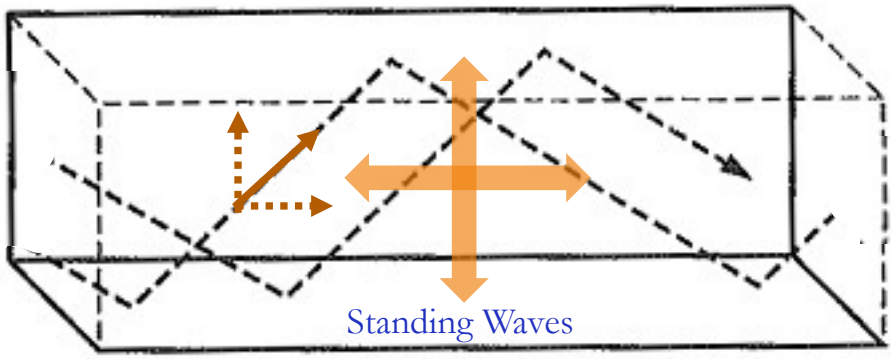




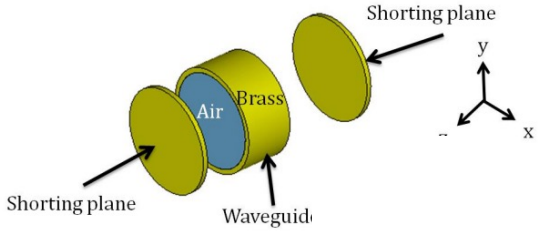
Introduction: Metallic Waveguides

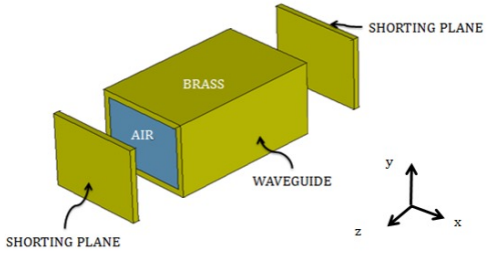
High energy and high frequency (microwave) EM waves can be transported for short distances using "waveguides" made of metal.

A metallic waveguide is a cavity with two opposite sides removes and one dimension extended.



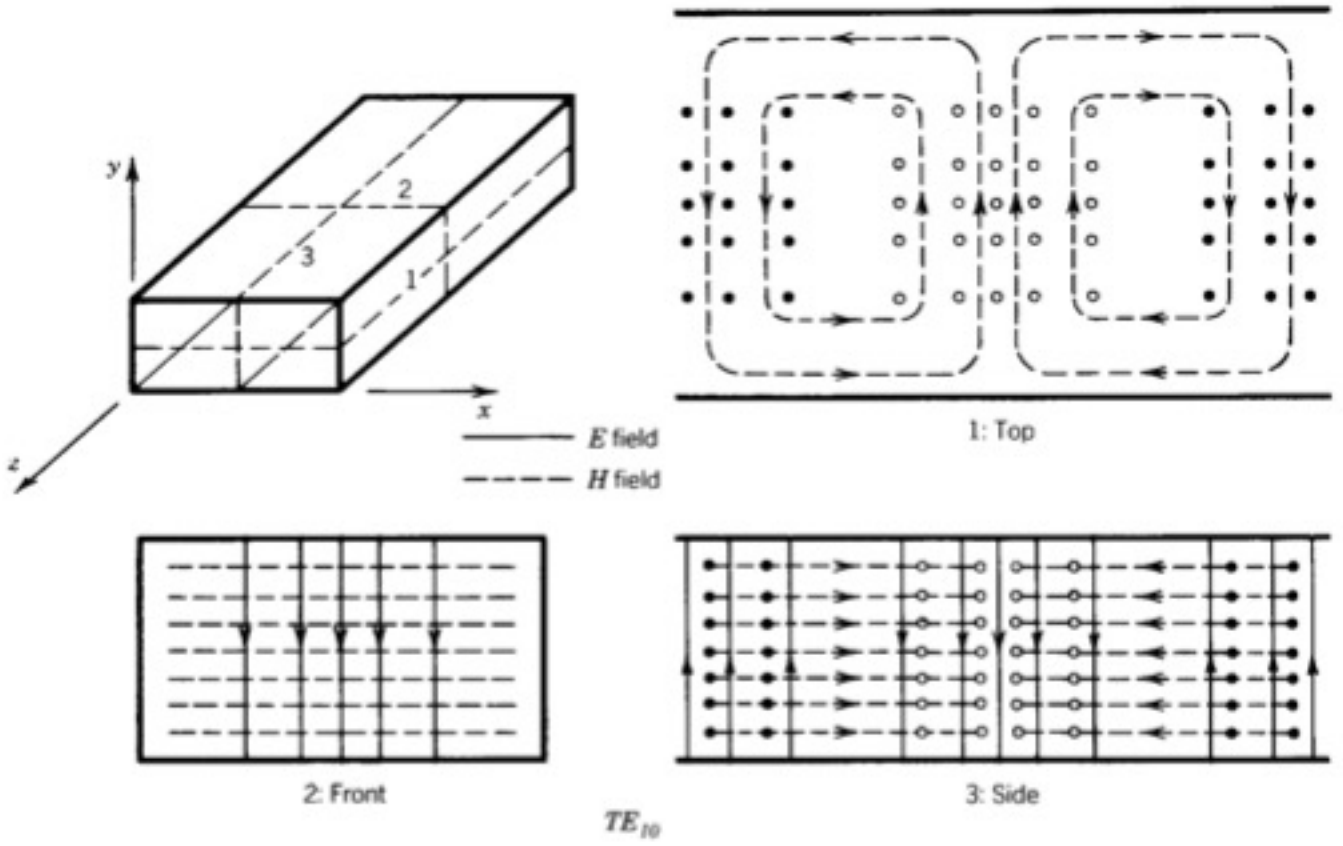




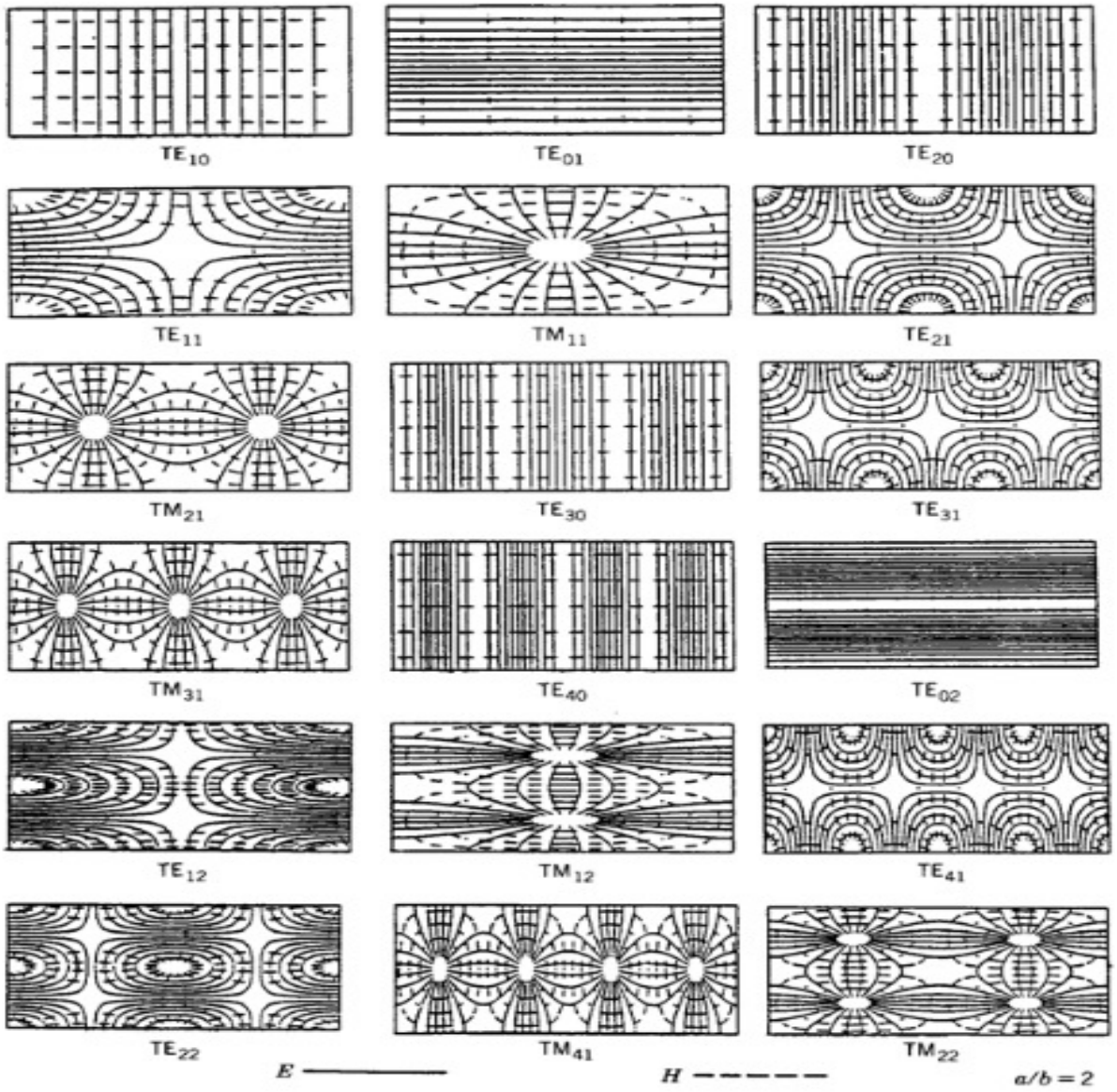


Modes in a Metallic Rectangular Waveguide

The concept of wave confinement here is based on perfect reflection over four metallic surfaces. The wave is allowed to propagate in one direction.



Modes in a Metallic Rectangular Waveguide



Wave Equation Solutions: TM Modes

Without going into complicated details, we can construct the required solutions by inspection.

1) TM Modes ($H_z = 0$)

$$E_z = (C_1 \cos\beta_x x + C_2 \sin\beta_x x) \cdot (C_3 \cos\beta_y y + C_4 \sin\beta_y y) \cdot e^{-j\beta_z z}$$

Apply boundary conditions, tangential fields=0 at $x = 0, a$ and $y = 0, b$.

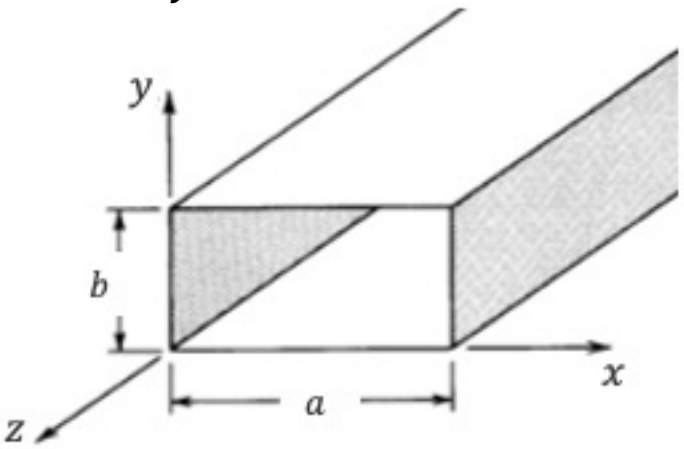
$$E_z = E_o \sin\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right) \cdot e^{-j\beta_z z}$$

$m = 1, 2, \cdots$ $\quad$ $n = 1, 2, \cdots$

$$\beta_x = \frac{m\pi}{a} \qquad \beta_y = \frac{n\pi}{b}$$

$$\beta^2 = \omega^2\varepsilon\mu = \beta_x^2 + \beta_y^2 + \beta_z^2$$

Mode number is TM_{mn}



Modes TM_{00} , TM_{01} and TM_{10} are not possible ($E_z = 0$)

Wave Equation Solutions: TM Modes

To find other field components, use Maxwell's equations.

$$E_z = E_o \sin\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right) \cdot e^{-j\beta_z z}$$

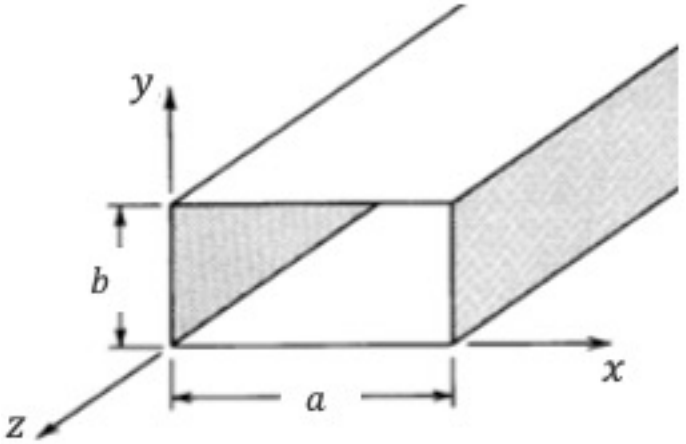
$$E_x = \frac{-j\beta_z}{\beta_{xy}^2}\left(\frac{m\pi}{a}\right) E_o \cdot \cos\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right) \cdot e^{-j\beta_z z}$$

$$E_y = \frac{-j\beta_z}{\beta_{xy}^2}\left(\frac{n\pi}{b}\right) E_o \cdot \sin\left(\frac{m\pi}{a}x\right) \cdot \cos\left(\frac{n\pi}{b}y\right) \cdot e^{-j\beta_z z}$$

$$H_x = \frac{j\omega\varepsilon}{\beta_{xy}^2}\left(\frac{n\pi}{b}\right) E_o \cdot \sin\left(\frac{m\pi}{a}x\right) \cdot \cos\left(\frac{n\pi}{b}y\right) \cdot e^{-j\beta_z z}$$

$$H_y = \frac{-j\omega\varepsilon}{\beta_{xy}^2}\left(\frac{m\pi}{a}\right) E_o \cdot \cos\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right) \cdot e^{-j\beta_z z}$$

$$H_z = 0$$



$$f_{mn} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

Cut-off (resonance) frequency for mode TM_{mn}

$$Z_{mn} = \eta\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}$$

Wave impedance for mode TM_{mn} (frequency dependent)

The dominant mode is the mode that will be excited first and has the lowest resonance frequency (TM_{11}).

Wave Equation Solutions: TE Modes

Without going into complicated details, we can construct the required solutions by inspection.

2) TE Modes ($E_z = 0$)

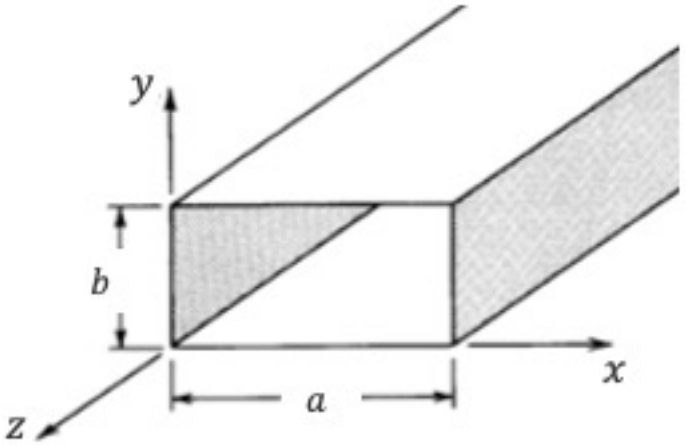
$$H_z = H_o \cos\left(\frac{m\pi}{a}x\right) \cdot \cos\left(\frac{n\pi}{b}y\right) \cdot e^{-j\beta_z z}$$

$$m = 0, 1, 2, \cdots \qquad n = 0, 1, 2, \cdots$$

$$\beta_x = \frac{m\pi}{a} \qquad \beta_y = \frac{n\pi}{b}$$

$$\beta^2 = \omega^2 \varepsilon \mu = \beta_x^2 + \beta_y^2 + \beta_z^2$$

Mode number is TE_{mn}



Modes TE_{00} is not possible

But modes TE_{m0}. and TE_{0n} are valid

Note: m and n cannot be both zero at the same time

Wave Equation Solutions: TE Modes

To find other field components, use Maxwell's equations.

$$H_z = H_o \cos\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta_z z}$$

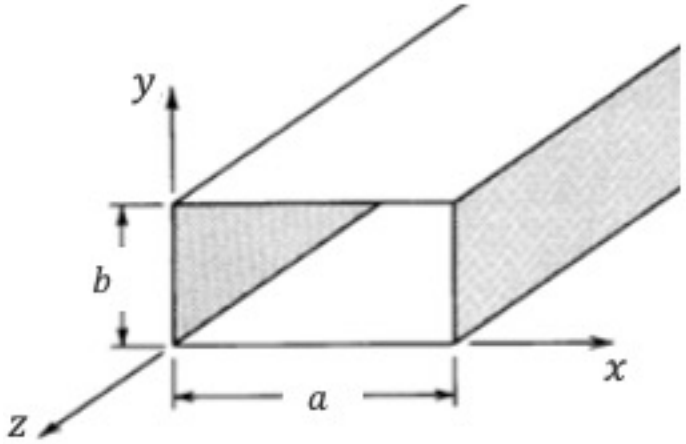
$$H_x = \frac{j\beta_z}{\beta_{xy}^2}\left(\frac{m\pi}{a}\right)H_o\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta_z z}$$

$$H_y = \frac{j\beta_z}{\beta_{xy}^2}\left(\frac{n\pi}{b}\right)H_o\cdot\cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta_z z}$$

$$E_x = \frac{j\omega\mu}{\beta_{xy}^2}\left(\frac{n\pi}{b}\right)H_o\cdot\cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta_z z}$$

$$E_y = \frac{-j\omega\mu}{\beta_{xy}^2}\left(\frac{m\pi}{a}\right)H_o\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta_z z}$$

$$E_z = 0$$



$$f_{mn} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}$$

Cut-off (resonance) frequency for mode TE_{mn}

$$Z_{mn} = \frac{\eta}{\sqrt{1-\left(\frac{f_{mn}}{f}\right)^2}}$$

Wave impedance for mode TE_{mn} (frequency dependent)

The dominant mode is the mode that will be excited first and has the lowest resonance frequency (TE_{10} or TE_{01}).

Example

Consider a metallic rectangular waveguide with dimensions 2.5 cm × 1 cm, filled with a dielectric of $\varepsilon_r = 4$ and $\sigma = 0$. Find the cutoff frequencies of all modes up to 15 GHz.

$$f_{mn} = \frac{c}{4}\sqrt{(40\,m)^2 + (100\,n)^2}$$

11 TE modes
4 TM mode

Dominant mode:
TE_{10}

Single mode operation between 3.0-6.0 GHz

TE_{mn}	f_{mn} (GHz)	TM_{mn}	f_{mn} (GHz)
10	3	11	8.078
01	7.5	21	9.6
11	8.078	31	11.72
02	15	41	14.14
20	6		
30	9		
40	12		
50	15		
21	9.6		
31	11.72		
41	14.14		

Q & A

Send me an email if you have any questions.